02 · Does the offer work better for some segments? — segment effects (pathmc)

July 6, 2026

1 02 · Does the offer work better for some segments? — segment effects (pathmc)

The decision. The email lifts spend *on average*. But should we send it to everyone, or does it really only move **high-value / highly-engaged** customers and barely touch the rest? If the segments genuinely differ we write segment-specific rules; if the "gap" is noise we shouldn't overfit our marketing to it — and we should know **what treating everyone as average costs us**.

We answer with a **structural model** and the **do-operator**: `pathmc` builds the causal graph, simulates `do(email=1)` vs `do(email=0)`, and reports the effect *per segment* and *as a smooth function of a continuous moderator*, with full posterior uncertainty — including a direct read on **whether the segments truly differ** (the posterior of the interaction).

7-step contract: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Customers are **high- or low-value** (`prior_value`) and also carry a continuous **`engagement`** score. We plant a real, layered moderation: the email lifts low-value customers by **€3** and high-value by **€12** at average engagement, **plus** a smooth **+€8 x engagement** slope on top. So the true per-customer effect is `tau = 3 + 9·is_high + 8·engagement` — a genuine, sizeable, and *continuous* heterogeneity. `prior_value` also drives spend directly (must be included); there's a background `trend`.

```
TRUE segment-average effects (at mean engagement): {'low': 7.0, 'high': 16.0}
TRUE effect spans €3.0 … €19.9 across customers
```

	email	prior_value	engagement	trend	spend	is_high
0	0.0	low	0.918802	58.466377	23.337758	0.0
1	0.0	low	0.651170	60.223782	19.057907	0.0
2	1.0	high	0.242720	53.530129	50.963256	1.0
3	0.0	low	0.119019	37.977148	12.868787	0.0
4	1.0	low	0.488445	61.807028	29.494100	0.0

1.2 3 · Identify — CATE via the do-operator

Estimand. The conditional effect $\tau(x) = \mathbb{E}[Y \mid do(\text{email}{=}1), x] - \mathbb{E}[Y \mid do(\text{email}{=}0), x]$. With a structural equation spend $= \beta_0 + \beta_1 e + \beta_2 v + \beta_3 (e{\cdot}v) + \beta_4 (e{\cdot}g) + \ldots$ the effect is $\tau = \beta_1 + \beta_3 v + \beta_4 g$ — **heterogeneity lives exactly in the moderation terms β_3, β_4.** No interaction => constant τ => `cate()` collapses to `ate()`. **Identification** is the usual unconfoundedness on email->spend (here email is randomized, so the admissible set is empty — the structural model just lets us *query* subgroup effects as functionals of the posterior). `pathmc` confirms the adjustment set:

```
admissible adjustment set(s) for email->spend: [set()]
identifiable: True
```

1.3 4 · Estimate — fit the structural model

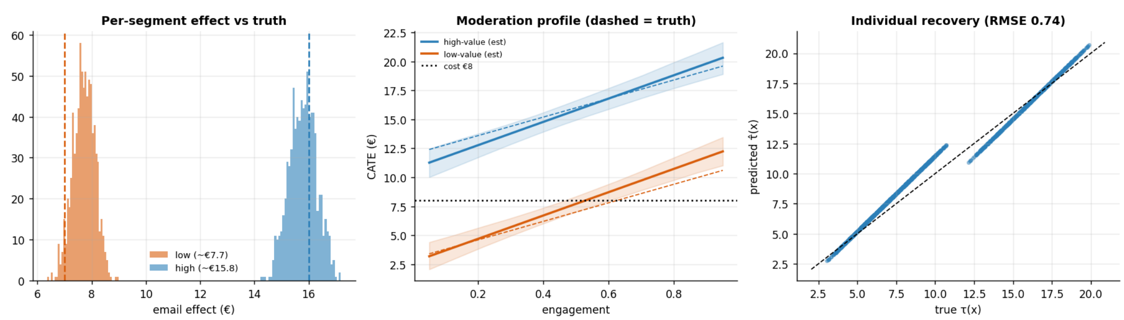
name	mean	sd	hdi_3%	hdi_97%
b_e	2.68	0.80	1.20	4.03
b_v	18.41	0.43	17.62	19.24
b_ev	8.08	0.61	6.96	9.24
b_eg	10.05	1.37	7.56	12.52
b_g	3.95	0.94	2.27	5.77
b_tr	0.29	0.02	0.27	0.32

1.4 5 · Validate — recover the effects, the gap, and the moderation profile

Four checks:

1. **Segment CATEs** vs the planted €3 / €12 — with credible intervals.
2. **Is the gap real?** the interaction posterior β_3; if it clearly excludes zero, the segments truly differ (not noise).
3. **The continuous moderation profile** — CATE as a smooth function of engagement, recovered vs the true `3 + 9·is_high + 8·engagement` line. This is the payoff of modelling the continuous moderator rather than binning.
4. **Do-operator distributions** — the posterior of spend under `do(email=1)` vs `do(email=0)` for a high-value customer, i.e. the intervention we're actually pricing.

```
ATE (pooled)      €10.80    [10.24147347 11.39268715]
CATE low  (eng .5) € 7.71  (true 7.0)    [6.96124494 8.45150667]
CATE high (eng .5) €15.79  (true 16.0)    [14.90499107 16.78183398]
Interaction beta(email:is_high) mean €8.08, P(>0) = 1.000 -> gap is REAL
```



1.5 6 · Decide, in euros — segment rules, and what pooling costs

Turn per-segment effects into **send rules** (target where P(effect > cost) is high), then quantify the money left on the table by **treating everyone as the pooled average** instead of segmenting — the business case for the whole exercise.

```
   segment  effect  net_of_cost  P(effect>cost) action
 low-value    7.71        -0.29            0.24   SKIP
high-value   15.79         7.79            1.00   SEND

Realised profit by policy:
  pooled (by ATE)    €3,682
  segment rules      €4,602
  individual tau_hat      €4,871
  oracle             €4,923

Moving from pooled to individual targeting is worth €1,189 (32% uplift).
```

1.6 7 · Caveats

- **Heterogeneity is only as good as the moderator you measured.** If the real driver of differential response isn't `is_high`/`engagement` (or a proxy), the interaction won't catch it.
- **Interaction != causing the moderator.** We're not claiming *making* someone high-value changes responsiveness — only that responsiveness *differs* across pre-existing segments.
- **Random slopes for many thin segments.** With dozens of small segments, replace the single interaction with partially-pooled random slopes so noisy segments borrow strength (see nb 03).

- **Same unconfoundedness caveat** — here email is randomized; observationally a hidden common cause of email and spend would bias every segment effect (see nb 05).